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DC-PPO for Joint User Association and Power Allocation in Dynamic Indoor Hybrid VLC/RF Networks

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***Abstract*—Hybrid visible light communication (VLC)/radio frequency (RF) networks have emerged as a promising next-generation networking technology that can provide high data rate and wide coverage while overcoming blockage problems of VLC and the limited available radio spectrum of RF. However, there has been limited research on the joint user association and power allocation problem in indoor hybrid VLC/RF networks with consideration of networking dynamics. To bridge the gap, we first formulate a joint user association and power allocation problem with the objective of maximizing the sum throughput of the hybrid networks with consideration of user mobility, time-varying blockages and interference between VLC access points (APs) and users, and practical capacity limitations of APs. Then we propose a distributed cooperative proximal policy optimization (DC-PPO) algorithm to solve the formulated mixed integer nonlinear programming problem, where each user is modeled as a hybrid agent that distributedly determines the user-AP association (discrete action) and the desired transmission power (continuous action) variables simultaneously based on local observations and limited information exchanged between neighboring nodes. The performance of our proposed method is compared with three baselines, including deep Q-network (DQN), max-power-max-signal-to-noise ratio (SNR), and random-power-max-SNR. Simulation results show that the proposed method is mobility and blockage-ware and outperforms the baseline solutions in terms of optimality, convergence, scalability, and robustness. Specifically, our proposed method can achieve 127% throughput improvement compared with that of DQN in 30-mobile-user scenario.**

***Index Terms*—Distributed PPO, user association and power allocation, hybrid VLC/RF networking, hybrid action space.**

I. Introduction

According to the report by Cisco [1], the number of wireless fidelity (Wi-Fi) hotspots is expected to increase by four times from 2018 to 2023. The increasingly dense deployment of indoor Wi-Fi hotspots has motivated the development of new technologies to address the traditional radio frequency (RF) spectrum shortage challenge. Visible light communication (VLC) technology, which utilizes the visible light spectrum for data transmission, has been emerging as a promising alternative to traditional RF networking systems for the following reasons [2]–[5]. First, it offers higher bandwidth and data transfer rates because it relies on a significant portion of the unregulated higher-frequency spectrum (ranging from 375 THz to 750 THz). Second, VLC is more secure than RF as lightwaves cannot penetrate through walls or other opaque objects, making it difficult for unauthorized individuals to intercept communication. Third, VLC is more energy-efficient as it uses low-power consumption LED lights to implement dual illumination and communication functionality. Finally, due to low electromagnetic interference, VLC can be used in areas where RF communication is restricted, such as hospitals and airplanes.

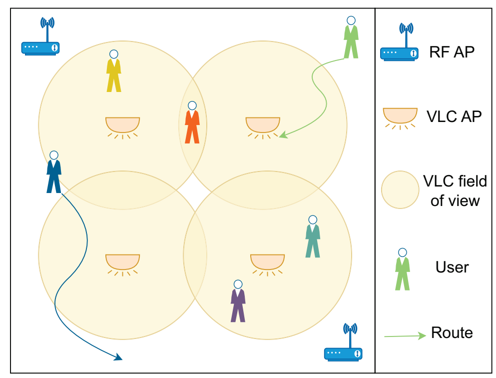

Fig. 1: Indoor hybrid VLC/RF networking scenario.

Although VLC has the above-mentioned advantages, it also has some limitations, including limited coverage area, and high sensitivity to blockage and alignment between transceivers. To overcome these limitations, a hybrid VLC/RF network [6] is proposed to combine the strengths of both technologies, i.e., the higher reliability and larger coverage of RF and the higher data rate and lower latency of VLC. Figure 1 presents a vision of indoor hybrid VLC/RF networks, where users can be served either by VLC or RF network. This resulting heterogeneous network (HetNet) presents several challenges to be addressed [7]. One of the biggest challenges is associating users with either VLC- or RF-based networks without interrupting concurrent communication. Another challenge is optimizing the resource allocation with consideration of the varying channel conditions, required data rate, interference, and power consumption to improve the network performance. Furthermore, these challenges become even more intricate when taking into account the dynamic nature of the networks, such as user mobility, and time-varying blockage for visible light line of sight (LoS) path.

Reinforcement learning (RL) has emerged as a robust solution for addressing complex, dynamically evolving problems,

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such as those encountered in wireless network configurations. This is because it learns optimal strategies through iterative trial-and-error interactions with a changing environment, thereby ensuring continuous adaptation and decision-making that aims to maximize long-term rewards. This methodology has been effectively applied to dynamic wireless network challenges [8], [9]. The joint user association and power allocation problem in hybrid VLC/RF networks is characterized by a hybrid action space that encompasses both discrete (user association) and continuous (power allocation) actions. Traditional RL algorithms, such as Q-learning and deep Q-networks (DQN), are predominantly tailored for discrete action spaces and often need to discretize continuous actions. However, this approach may lead to suboptimal performance due to the potential misalignment of the optimal action with the discretized alternatives. To address this challenge, RL algorithms like deep deterministic policy gradient and proximal policy optimization (PPO) have been developed, to handle continuous action spaces. While these algorithms have the potential to deal with discrete action spaces, however, addressing hybrid action spaces remains inherently more complex compared to environments with solely discrete or continuous actions. In [10], the authors propose a double-agent policy generation model, aimed at separately addressing discrete and continuous actions. However, this approach faces significant challenges in terms of learning efficiency and increased complexity. In contrast, we propose an innovative reinforcement learning-based framework specifically designed to address the joint discrete user association and continuous power allocation challenge. This framework employs a single, hybrid agent capable of generating both action types. This approach effectively avoids the communication difficulties and convergence issues that are common in multi-agent models, thereby simplifying the learning process and reducing the complexity of the overall system.

In our previous work [11], we propose a centralized RL based method to solve the user-AP association problem. While this method outperforms state of art, it is observed that its time complexity escalates progressively with the increase in the number of users. To address this challenge, in this paper, we propose a distributed cooperative algorithm based on proximal policy optimization to solve the joint user association and power allocation problem in indoor hybrid VLC/RF networks with the objective of maximizing the sum throughput of the network. Different from most existing works, where they are either focused on static networking scenarios without consideration of user mobility and time-varying blockage between users and VLC access points (APs) [12]–[14] or dynamic networks without practical channel capacity limitation [15] or centralized learning solution algorithms [16], [17], we formulate the sum-rate optimization problem by jointly considering user mobility, blockage, and practical capacity limitations. Moreover, our proposed distributed learning algorithm runs on each user device and only requires limited cooperation and information exchange among neighboring wireless devices, thus avoiding the collection of real-time global information of the network and reducing the learning model complexity [9], [18], [19].

Our paper makes the following novel contributions:

- *Joint user association and power allocation.* We first present a mathematical formulation of the joint user association and power allocation problem, where the objective is to maximize the sum throughput of the indoor hybrid VLC/RF networks by considering the user mobility, time-varying blockage between VLC APs and users, as well as interference among VLC APs and users. It is shown that the resulting problem is a mixed integer nonlinear programming (MINLP) problem.
- *Distributed cooperative PPO solution algorithm.* To solve the formulated MINLP problem, we propose a distributed proximal policy optimization algorithm, a deep reinforcement learning algorithm based on actor-critic architecture. The algorithm runs on each user agent, thus eliminating the problem of action spaces growing exponentially larger as the numbers of user and AP increase, and reducing the information exchange requirements as in the centralized method. Moreover, PPO is capable of directly handling the hybrid action space [20], i.e., discrete user association variable and continuous power variable in our proposed problem.
- *Performance evaluation.* We demonstrate the effectiveness of our proposed method by comparing simulation results with existing methods in hybrid VLC/RF networks, including max-power-max-signal-to-noise ratio (SNR), random-power-max-SNR, and deep Q-learning network. The results show that our proposed method achieves significant improvements in terms of convergence speed, network throughput, user satisfaction, and robustness in both static and dynamic scenarios with varying interference and blockage.

The remaining sections of this paper are organized as follows. The related works are reviewed in Section II. In Section III, we describe the VLC and RF channel model, as well as the dynamic nature of the network. The proposed joint problem is formulated in Section IV. Then, Section V describes in detail the proposed distributed cooperative PPO (DC-PPO) method. We then analyze the computational complexity and discuss the simulation results in Section VI. Finally, Section VII concludes the paper.

II. Related Work

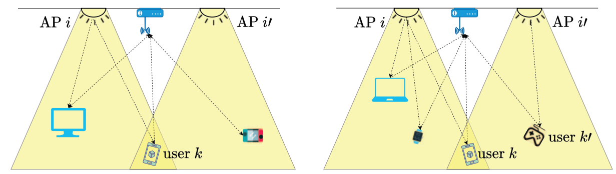
In recent years, there has been a growing interest in improving the network throughput in indoor hybrid VLC/RF networks. Various techniques such as fuzzy logic [21], optimization-based method [22], [23], game theory [24], Q-learning [25], trust region policy optimization [16] have been proposed to tackle the user-AP association problem. There are also some other works that focus on the power allocation problem [26], [27]. However, focusing only on one single problem cannot achieve better network performance because user association and power allocation are coupled. For example, the user association can be used to improve the performance of the power allocation to better manage the interference among users, and vice versa.

To bridge this gap, a few works have started to study the joint user association and power allocation problem in hybrid VLC/RF networks [12]–[14]. The authors in [12] propose a joint load balancing and power allocation scheme to maximize the achievable data rate in static hybrid VLC/RF networks and derive an iterative solution algorithm to distribute users to APs as well as allocate the power of the APs to their served users with consideration of different blockage occurrence probabilities. In [13], an unsupervised deep neural network (DNN) based learning algorithm is proposed to solve the user association and power allocation problems simultaneously, where time complexity and network throughput are evaluated. [14] reformulates the joint user-AP association and power allocation problem into the college admission model and then proposes a distributed and low-complexity solution algorithm based on matching theory. However, these aforementioned works lack generality and practicality pertaining to no consideration of either user mobility, time-varying blockage in real indoor environments, or network capacity limitation.

There do have some literature on heterogeneous RF networks, such as heterogeneous cellular networks, focusing on joint user association and power allocation problem [15], [19], [28]. However, they cannot be directly applied to indoor hybrid VLC/RF networks because of the unique characteristics of VLC networks, such as limited coverage range due to limited field of view (FoV) and susceptibility to blockage. To the best of our knowledge, we formulate for the first time a joint user association and power allocation problem in an indoor hybrid VLC/RF network with comprehensive consideration of practical factors including dynamic networking topologies with mobile users, time-varying interference and blockage situations, and channel capacity limitations. This is also the first time, we propose a distributed cooperative PPO based algorithm to solve the resulting MINLP problem with the hybrid continuous and discrete action space in a computation-efficient way.

III. System Model

We consider an indoor hybrid VLC/RF network as illustrated in Fig. 1, where a set of VLC APs (denoted by $\mathcal{I}$ = $\{1,\ 2,\ ...,\ i,\ ...,\ |I|\}$) and RF APs ($\mathcal{J}$ = $\{1,\ 2,\ ...,\ j,\ ...,\ |J|\}$) provide down-link access to a set of users (denoted by $\mathcal{K}$ = $\{1,\ 2,\ ...,\ k,\ ...,\ |K|\}$). In the remainder of the paper, $|\cdot|$ represents the cardinality of the set. We assume that the VLC APs are installed on the ceiling at pre-defined locations, straightly facing downwards. Without loss of generality, a time division multiple access scheme is adopted at each AP, which can be easily changed to other multiple access methods. The capacity limitations of VLC and RF APs are defined as the maximum user number they can serve, denoted as N_i and N_j respectively. We also consider a single-home strategy for the users in the hybrid indoor VLC/RF networks, which means that each user can be served by at most one AP at a time.



(a) Interference-free scenario (b) Interference-prone scenario

Fig. 2: Indoor VLC networking interference scenarios.

A. VLC Channel Model

In this work, we consider an intensity modulation and direct detection (IM/DD) channel model. For simplicity, we also only consider the LoS path for VLC networks in the proposed model. However, this can be easily extended to combined LoS and non-LoS VLC networking scenarios. Then, we can calculate the LoS channel gain $H_{i,k}$ between VLC AP i and user k based on Lambertian radiation pattern [29] as:

$$H_{i,k}=\begin{cases}\frac{(m+1)A_{PD}}{2\pi d_{i,k}^2}\cos^m(\phi_{i,k})T_s(\psi_{i,k})g(\psi_{i,k})\cos(\psi_{i,k}) & \\ \qquad\qquad\qquad\qquad 0\le\psi_{i,k}\le\Psi_{1/2}, & \\ 0 \qquad\qquad\qquad\qquad otherwise, & \end{cases} \tag{1}$$

where m is the order of the Lambertian emission index and is calculated as $m=-\ln 2/\ln(\cos\Psi_{1/2})$, with $\Psi_{1/2}$ representing the semi-angle at half illumination power of a LED. A_{PD} represents the physical area of the receiver's photo diode (PD). $d_{i,k}$ denotes the distance between VLC AP i and user k. $\phi_{i,k}$ and $\psi_{i,k}$ are the irradiance angle and incidence angle, respectively. $T_s(\psi_{i,k})$ is the gain of the optical filter. $g(\psi_{i,k})$ is the optical concentrator gain, given as:

$$g(\psi_{i,k})=\begin{cases}\frac{n^2}{\sin^2\Psi_{1/2}} & 0\le\psi_{i,k}\le\Psi_{1/2}, \\ 0 & otherwise,\end{cases} \tag{2}$$

with n denoting the refractive index.

In the considered scenario, the deployed multiple VLC APs can be used to transmit data, provide illumination, or both. This motivates us to study the optical interference types in indoor VLC/RF networks. We for the first time conclude two cases as shown in Fig. 2 and discussed in the following.

Interference-free case: As illustrated in Fig. 2(a), a user k is located in the overlapping area of the FoV of VLC AP i and VLC AP i'. If user k chooses to connect to AP i, AP i' does not serve any users and thus solely operating in illumination mode. In this case, AP i' will not cause any optical interference to user k since it emits constant-intensity illumination light waves which can be considered as white noise at the receiver side [30]. Therefore, only the signal-to-noise ratio (SNR) $\delta_{i,k}$ between user k and VLC AP i should be used to calculate the throughput, given as:

$$\delta_{i,k}=\frac{(\kappa H_{i,k}p_{i,k})^2}{\mathcal{N}_{i,k}B_{i,k}}, \tag{3}$$

where κ is optical to electric conversion efficiency of the PD. $p_{i,k}$ denotes the transmit power between VLC AP i and user k. $\mathcal{N}_{i,k}$ and $B_{i,k}$ represent the noise power spectral density and the modulation bandwidth, respectively.

Interference-prone case: Different from *interference-free* *case*, if VLC AP i' is also concurrently serving user k' while VLC AP i is serving user k as shown in Fig. 2(b), VLC AP i' will then introduce interference to user k. Thus, signal-to-interference-plus-noise ratio (SINR) $\gamma_{i,k}$ between user k and VLC AP i should be considered when calculating the capacity of user k, which is calculated as:

$$\gamma_{i,k} = \frac{(\kappa H_{i,k} p_{i,k})^2}{\mathcal{N}_{i,k} B_{i,k} + \sum\limits_{i' \neq i, k' \neq k} (\kappa H_{i',k} p_{i',k'})^2}, \tag{4}$$

where the $p_{i',k'}$ represents the transmit power between interfering AP i' and its corresponding connected user k'. $H_{i',k}$ is the channel gain between the overlapping user k and the interfering VLC AP i'.

Then we can calculate the data rate of user k. However, the non-negative nature of optical signals in visible light communication renders the Shannon capacity inapplicable. Therefore, a lower bound on the achievable data rate for user k of VLC AP i in [31] is used. Considering the two interference-free and interference-prone cases, the achievable data rate is defined as:

$$C_{i,k} = \begin{cases} B_{i,k} \log_2(1 + \frac{e}{2\pi}\delta_{i,k}) & \text{interference-free case,} \\ B_{i,k} \log_2(1 + \frac{e}{2\pi}\gamma_{i,k}) & \text{interference-prone case.} \end{cases} \tag{5}$$

B. RF Channel Model

The channel gain between RF AP j and user k can be expressed as [32]:

$$H_{j,k} = \sqrt{10^{\frac{-L(d_{j,k})}{10}}} \left(\sqrt{\frac{K}{1+K}} e^{j\phi} + \sqrt{\frac{1}{1+K}} X\right), \tag{6}$$

where K is the Rician factor for RF links [32], $K = 0$ if distance $d_{j,k}$ is greater than break point d_{BP}, otherwise $K = 1$. ϕ is the angle of arrival or departure of the RF LoS component. $X \sim \mathcal{CN}(0, 1)$ represents the fading channel of the scattered path. The path loss function, $L(\cdot)$, is determined based on the distance between the RF AP j and the user k, i.e., $d_{j,k}$. $L(d_{j,k})$ is calculated as :

$$L(d_{j,k}) = \begin{cases} L_F(d_{j,k}) + X_S & d_{j,k} \leq d_{BP}, \\ L_F(d_{j,k}) + X_S + 35\log_{10}(\frac{d_{j,k}}{d_{BP}}) & d_{j,k} > d_{BP}, \end{cases} \tag{7}$$

with L_F referring to the free space loss, calculated as $L_F(d_{j,k}) = 20\log_{10}(d_{j,k}) + 20\log_{10}(f_c) - 147.5$, where f_c denotes the central carrier frequency. While X_S denotes the shadow fading loss, following a Gaussian distribution with a mean of zero and a 3 dB deviation if $d_{j,k} > d_{BP}$, otherwise the deviation being 5 dB. RF APs can operate in different frequency channels to avoid interference in a multi-RF-AP network. In this case, only the SNR at user k from RF AP j is considered and given as:

$$\delta_{j,k} = \frac{|H_{j,k}|^2 p_{j,k}}{\mathcal{N}_{j,k} B_{j,k}}, \tag{8}$$

where $\mathcal{N}_{j,k}$ and $B_{j,k}$ stand for power spectral density of additive white Gaussian noise and bandwidth between RF AP j and user k, respectively. And $p_{j,k}$ is the allocated power from the RF AP j for user k. Otherwise, the SINR should be used and calculated as

$$\gamma_{j,k} = \frac{|H_{j,k}|^2 p_{j,k}}{\mathcal{N}_{j,k} B_{j,k} + \sum\limits_{j' \neq j, k' \neq k} |H_{j',k}|^2 p_{j',k'}}. \tag{9}$$

Finally, We can calculate the achievable data rate $C_{j,k}$ in the RF network as:

$$C_{j,k} = \begin{cases} B_{j,k} \log_2(1 + \delta j, k) & \text{operate in different frequency channels,} \\ B_{j,k} \log_2(1 + \gamma_{j,k}) & \text{otherwise.} \end{cases} \tag{10}$$

C. Mobility Model

To simulate user mobility, a random waypoint model (RWP) is adopted [33], which is commonly utilized in wireless network simulations [16], [34]. In this model, each user moves randomly within a predefined indoor space with a randomly selected speed and destination. Once the destination is reached, each user selects to either pause for a preset time or continue moving with a newly generated random speed and destination. This work can be easily extended to other mobility models.

D. Blockage Model

Because of the susceptibility of VLC AP to blockage caused by obstacles, the LoS path between LED transmitters and PD receivers may not be always available. Therefore, we also design a blockage scheme to simulate the time-varying blockage state in the proposed indoor scenarios. To be specific, each user k will be assigned a randomly generated blockage probability $\Pr(i, k)$ between itself and VLC AP i. Let $\rho_{i,k}$ represent the corresponding binary blockage indicator. The binary indicator will be set to 1 when the blockage probability $\Pr(i, k)$ is greater than a given threshold $\Pr_{th}$, indicating an unobstructed LoS path, otherwise 0, indicating a blocked path. It is worth noting that the blockage only occurs when the VLC AP and user are within each other's FoV, and both $\Pr(i, k)$ and $\rho_{i,k}$ are time-varying parameters.

IV. Problem Formulation

As discussed in Sec. II, user association and power control play a pivotal role in hybrid VLC/RF networks. On the one hand, interference is directly affected by user association decisions and various allocated power levels. On the other hand, user association and power control strategies are the most straightforward way to solve the load balancing problem among different APs. Therefore, in this section, we mathematically formulate a network control optimization problem of the indoor VLC/RF network with the objective of maximizing the sum throughput by jointly determining the user association vector and power allocation vector with consideration of time-varying user positions and blockage, different interference cases, as well as capacity limitations of APs.

Let $x_{i,k}$ and $x_{j,k}$ represent the *binary* association variable between VLC AP i and user k, and RF AP j and user k, respectively. Then, the association vector for user k can be denoted as $\mathbf{x}_k = [x_{i,k},\ x_{j,k}]$, where $x_{i,k} = 1$ or $x_{j,k} = 1$ indicates that the user k is associated with VLC AP i or RF AP j, otherwise $x_{i,k} = 0$ or $x_{j,k} = 0$. Since we consider a single-home association strategy, thus $\mathbf{1}^T \mathbf{x}_k \leq 1, \forall k \in \mathcal{K}$. We then can obtain the user association matrix of the network as $\mathbf{x} = [\mathbf{x}_1, ..., \mathbf{x}_k, ..., \mathbf{x}_{|K|}]$. The *continuous* allocated power vector of user k and matrix of the network are denoted as $\mathbf{p}_k = [p_{i,k},\ p_{j,k}]$ and $\mathbf{p} = [\mathbf{p}_1, ..., \mathbf{p}_k, ..., \mathbf{p}_{|K|}]$, respectively.

Based on VLC and RF channel models described in section III-A and III-B, the achievable data rate of user k is given as:

$$C_k = \sum_{i=1}^{\mathcal{I}} x_{i,k}\rho_{i,k}C_{i,k} + \sum_{j=1}^{\mathcal{J}} x_{j,k}\rho_{j,k}C_{j,k}, \tag{11}$$

and the achievable sum network throughput of the hybrid VLC/RF network can be expressed as:

$$C_{sum} = \sum_{k=1}^{\mathcal{K}} C_k. \tag{12}$$

Finally, the joint user association and power allocation optimization problem can be formulated as:

$$\text{Problem 1: } \underset{\mathbf{x},\mathbf{p}}{\text{Maximize}} \quad f = C_{sum} \tag{13}$$

$$\begin{aligned}
\text{Subject to:} \quad & x_{i,k} \in \{0,1\}, \forall i \in \mathcal{I}, \forall k \in \mathcal{K}, && (14)\\
& x_{j,k} \in \{0,1\}, \forall j \in \mathcal{J}, \forall k \in \mathcal{K}, && (15)\\
& \mathbf{1}^T\mathbf{x}_k \leq 1, \forall k \in \mathcal{K}, && (16)\\
& \sum_{k=1}^{\mathcal{K}} x_{i,k} \leq N_i, \forall i \in \mathcal{I}, \forall k \in \mathcal{K}, && (17)\\
& \sum_{k=1}^{\mathcal{K}} x_{j,k} \leq N_j, \forall j \in \mathcal{J}, \forall k \in \mathcal{K}, && (18)\\
& 0 \leq p_{i,k} \leq P_i, \forall i \in \mathcal{I}, \forall k \in \mathcal{K}, && (19)\\
& 0 \leq p_{j,k} \leq P_j, \forall j \in \mathcal{J}, \forall k \in \mathcal{K}, && (20)
\end{aligned}$$

where constraints (14) and (15) indicate that the user association variables are binary. Moreover, constraint (16) guarantees that every user can only connect to at most one VLC or RF AP. Constraints (17) and (18) ensure the capacity limitations for VLC and RF AP, respectively. Constraints (19) and (20) impose a non-negative power allocation and a power budget limit for the VLC and RF APs, respectively.

Because of the non-convexity of C_k with respect to $\gamma_{i,k}$ and $\gamma_{j,k}$ and the binary user association variables, the formulated problem is a MINLP problem, for which there exists no known traditional optimization based algorithms that guarantee the optimal solution in polynomial time.

V. Distributed Cooperative Proximal Policy Optimization Solution Algorithm

Reinforcement learning has been adopted as a ubiquitous method to solving non-convex optimization problems in wireless communication [13], [15], [16], [19], [25], [28]. Currently, most existing learning solutions for wireless networks rely on centralizing the training that requires global network information, thus resulting in significant overhead and computational complexity, which dramatically increases as the number of APs and users increases. Moreover, the widely adopted value-based RL methods such as Q-learning and DQN are incapable to find the optimal solution for problems with hybrid action space as in the proposed Problem 1, because the selected actions will be infinite for continuous action space [35].

Therefore, we propose a distributed cooperative proximal policy optimization method because (i) DC-PPO is based on an actor-critic RL framework that combines the actor network (i.e., policy) and the critic network (i.e., estimated value function) [36] to improve sample efficiency of policy-based methods and convergence speed when learning continuous-valued policies of value-based methods [37]; (ii) Unlike the widely used DQN or DDPG methods, which are designed to handle either discrete action space or continuous action space, respectively, DC-PPO is able to tackle hybrid discrete and continuous actions simultaneously. This is because the actor network in DC-PPO can output a Gaussian distribution, where an activation function such as *tanh* or *softmax* can be used to generate continuous actions or discrete actions, respectively [38]. (iii) The proposed DC-PPO distributedly runs on each user agent, where limited cooperation and information exchange among neighboring user agents are needed instead of global network information. Therefore, it can not only handle hybrid discrete and continuous action space to output the optimal or near-optimal solutions through trial-and-error interactions with the environment, but also reduce latency caused by instantaneous global information collection and computational complexity by enabling distributed learning at each user compared with complex centralized learning, thus being scalable. Next, we discuss the proposed DC-PPO method in detail.

A. Overview of DC-PPO

Each user acts as a distributed DC-PPO agent and its process of joint user association and power allocation problem is modeled as a Markov process. At time t, each user agent chooses an action a_t based on the stochastic policy $\pi_\theta(a_t|s_t)$ at the current state s_t in the environment, where θ denotes the parameters of the policy. After which, the environment generates an immediate reward r_t to each agent and then transits to the next state s_{t+1}. The user agents will store this experience in memory as a tuple $\{s_t, a_t, r_{t+1}, s_{t+1}\}$ for policy updates. The objective of each agent is to find an optimal policy π^* that maximizes the expected accumulated discounted reward, denoted as $\mathbb{E}[\sum_{t=0}^{\infty}\beta^t r_t]$, with $\beta \in (0,1]$ being a discount factor.

Specifically, the DC-PPO agent is composed of two networks: the actor network and the critic network. The actor network is used to define parameterized policy and generate actions according to the observed environment state, while the role of the critic network is to evaluate and criticize the current policy by processing the rewards received from the environment. After the value function approximation and its parameters are updated by the critic, the actor then uses the corresponding output from the critic network to update its policy parameters. One important feature of DC-PPO is to learn the optimal stochastic policy by minimizing a clipped surrogate objective as [38]:

$$L_t^{CLIP}(\theta) = \hat{\mathbb{E}}_t[min(r_t(\theta)\hat{A}_t, clip(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)], \tag{21}$$

with $\hat{A}_t$ being an estimator of the advantage function, and ϵ denoting a clipping hyperparameter. $r_t(\theta)$ represents the probability ratio, given as $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$, with $\pi_{\theta_{\text{old}}}$ denoting the old policy. This function constrains policy updates to a specified range, preventing large changes that could lead to instability or suboptimal policies. Moreover, by combining the clipped surrogate objective function (21) with the estimated value function from the critic network, the DC-PPO algorithm is able to make efficient use of collected samples, resulting in more effective learning and faster convergence.

Algorithm 1 Mobility and blockage aware DC-PPO for joint user association and power allocation problem in indoor hybrid VLC/RF networks

Input : Initialize $|K|$ agents, and initial state $\boldsymbol{s}(0)$
Output: Optimal policy $\boldsymbol{\pi}^*$
while *episode less than max episode* **do**
  while *timestep less than update timestep* **do**
    for *each agent* **do**
      Select action $a_k(t)$ according to policy π_k
    end
    Environment calculates received data rate $C_k(t)$ for every agent and sum network throughput $C_{sum}(t)$ based on all action $\mathbf{a}(t)$
    for *each agent* **do**
      Calculate local reward based on (22)
      Observe new state $s_k(t+1)$ based on local observations and previous experience
      Store experience into memory
    end
  end
  for *each agent* **do**
    Update policy π_k with the clipped surrogate objective (21)
  end
  Clear memory
  timestep = 0
end

B. Algorithm Design of DC-PPO

In the proposed DC-PPO framework, we model users as agents and assign them a shared objective of maximizing the sum throughput of the indoor hybrid VLC/RF networks. Each user agent executes the proposed network control algorithm to decide to connect to either one of the VLC or RF APs and requests specific transmission power from the associated AP. For agent k, the actions, states, policy, and reward are specifically defined as follows:

Actions: Let $\mathcal{A}$ denotes a set of actions, where each action includes discrete user association and continuous power allocation variables. At time slot t, the action of agent k is denoted as $a_k(t) = [\mathbf{x}_k(t), \mathbf{p}_k(t)]$, where $a_k(t) \in \mathcal{A}$ and $k \in \mathcal{K}$.

States: The set of states, denoted as $\mathcal{S}$, including the received SNR $\boldsymbol{\delta}_k(t)$, the required data rate $C_k^{req}(t)$, the LoS blockage status $block_k(t)$, the previously selected action $a_k(t-1)$, the received data rate $C_k(t-1)$ at time $t-1$, and network throughput $C_{sum}(t-1)$ inferred from locally collected data rate information from neighboring nodes at time $t-1$. This information enables cooperation of the actions of all agents, allowing each agent to comprehend how their actions will impact the overall objective. The state $s_k(t)$ of agent k at time t is represented as:

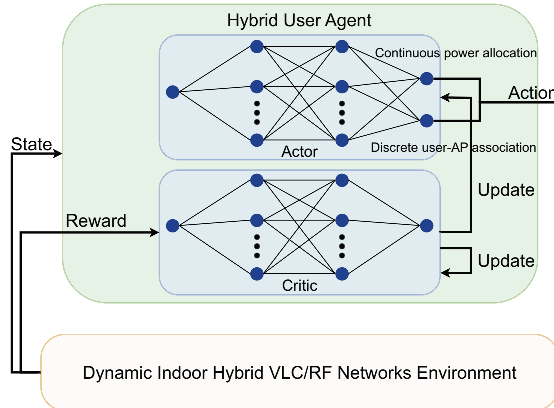


Fig. 3: Hybrid DC-PPO Agent Model.

$$s_k(t) = [\boldsymbol{\delta}_k(t), C_k^{req}(t), block_k(t), a_k(t-1), C_k(t-1), C_{sum}(t-1)],$$

with $\boldsymbol{\delta}_k(t) = \{\delta_{i,k}(t), \delta_{j,k}(t), \forall i \in \mathcal{I}, \forall j \in \mathcal{J}\}$, $s_k(t) \in \mathcal{S}$, and $k \in \mathcal{K}$.

Policy: A stochastic policy $\pi_{\theta,k}$ is adopted, which is the probability of choosing action $a_k(t)$ given state $s_k(t)$. Formally, $\pi_{\theta,k}(a_k(t)|s_k(t)) = \Pr(a_k(t)|s_k(t))$.

Reward: A weighted reward function is used to leverage the effect of individual achievable data rate and network sum throughput on the learning performance, where the network throughput $C_{sum}(t)$, the received data rate $C_k(t)$, and a penalty for violating the required data rate constraint are considered. The immediate reward $r_k(t)$ of agent k at time t is defined as:

$$r_k(t) = w_1 C_{sum}(t) + w_2 C_k(t) + w_3 (C_k(t) - C_k^{req}(t)), \tag{22}$$

where w_1, w_2, w_3 are the tunable weighted parameters determined empirically. Then, the discounted accumulated reward can be expressed as $R_k(t) = \sum_{t=0}^{T} \beta^t r_k(t)$.

The hybrid DC-PPO agent model is illustrated in Fig. 3. The dynamic indoor hybrid VLC/RF networks environment sends the current state to both actor and critic networks, then the actor network decides on both continuous power allocation and discrete user-AP association actions simultaneously based on its current policy, which is a strategy it has learned for making decisions. Upon executing the action, the critic network receives feedback from the environment in the form of a reward, which indicates how beneficial the action was towards achieving its goals. This reward feedback is then assessed by the critic network, and the outcome of this evaluation is utilized to update and improve both the actor and critic network, enhancing its decision-making over time. The detailed procedure of the proposed DC-PPO algorithm is explained in Algorithm 1.

VI. Performance Evaluation

A. Simulation setup

We consider an indoor room with dimension $8 \times 8 \times 3$ m^3, where $|\mathcal{J}| = 1$ RF AP is installed at the center of the ceiling covering the entire room, and $|\mathcal{I}| = \{4, 6\}$ VLC APs are installed at predefined positions on the ceiling to illuminate the room. Please note that based on the problem formulation discussed in Sec. VI, it is easy to extend the number of VLC or RF APs to any number. Users are randomly distributed within the defined room with moving speeds ranging from 0.1 m/s to 1 m/s, where users are assumed to be static within 10 ms, which is justified as the maximum distance a user can cover within a 10 ms time slot is 0.01 m [34]. The parameters used for hybrid VLC/RF networks are summarized in Table I and Table II. The RL parameters used in the simulation are summarized in Table III. We conduct our experiments on a 2021 MacBook Pro equipped with an Apple M1 chip, 16 GB of RAM, and a 500 GB SSD. The computer is running macOS Monterey version 12.5.1, and we use Python version 3.10.0, and PyTorch version 1.12.1 for deep reinforcement learning implementation.

B. Compared Methods

To evaluate the effectiveness of our proposed algorithm, we compare it with three state-of-the-art methods in the field of hybrid VLC/RF networks, including DQN, max-power-max-SNR (MPMS), and random-power-max-SNR (RPMS).

DQN: DQN is a widely adopted RL method in wireless networks for tasks such as user-AP association [18], [19], power allocation [9], and channel selection [39]. We use DQN as a baseline RL method to be compared with the proposed DC-PPO method for performance analysis in terms of the convergence speed and the achievable sum network throughput. For a fair comparison, DQN is also implemented in a distributed manner. Since DQN is only capable of handling discrete actions, we discretize the power action space into N_D levels, where N_D can be determined based on the tradeoff between the performance and the computational complexity, i.e., the larger N_D, the better performance but the higher computational complexity, vice versa.

TABLE I: VLC network parameters

Parameter	Value
Physical area of a PD (A_{PD})	1 cm^2
Gain of the optical filter ($T_s(\psi_{i,k})$)	1.0
Receiver FoV semi-angle ($\Psi_{1/2}$)	90 degree
Refractive Index (n)	1.5
Half-intensity radiation angle ($\Phi_{1/2}$)	60 degree
Detector responsivity (κ)	0.53 A/W
Maximum transmit optical power per VLC AP (P_i)	3 Watt
Noise power spectral density ($\mathcal{N}_i$)	10^{-21}A^2/Hz
Bandwidth per VLC AP (B_i)	40 MHz

TABLE II: RF network parameters

Parameter	Value
Breakpoint distance (d_{BP})	5 m
Central carrier frequency (f_c)	2.4 GHz
The angle of arrival/departure of LoS (ϕ)	45 degree
Maximum transmit power per RF AP (P_j)	0.1 Watt
Noise power spectral density ($\mathcal{N}_j$)	4.002x10^{-17}W/Hz
Bandwidth per RF channel (B_j)	10 MHz

MPMS: As the name suggests, the MPMS method lets users require the maximum transmission power from the VLC or RF APs and connect to the AP offering the highest SNR.

RPMS: RPMS method assigns random power within the power budget for all user-AP pairs. After that, the users then select the AP with the highest SNR to be associated. To make an equitable comparison with our proposed DC-PPO method, both MPMS and RPMS methods will be directly provided blockage information between users and VLC APs.

C. Complexity Analysis

In this subsection, we analyze the computational complexity of our proposed DC-PPO method and the three baseline methods mentioned above. It is worth noting that while the exhaustive search method can guarantee a global optimum, we don't use it since is not feasible for our formulated problem due to the following two reasons. First, the continuous power allocation variables will result in an infinite number of possible solution candidates, which cannot be handled by the brute-force exhaustive search approach. Second, even if we discretize the continuous variable into N_D levels, the complexity becomes $\mathcal{O}(((|\mathcal{I}|+|\mathcal{J}|)N_D)^{|K|})$, where $|\mathcal{I}|+|\mathcal{J}|$ is the number of APs. The computational complexity grows exponentially with an increasing number of users, making the implementation of the exhaustive method impossible on the personal computer device.

DC-PPO: Let N_H denote the number of hidden neurons in the hidden layer, and there are two hidden layers for both the proposed method and the DQN baseline. Considering the hybrid discrete and continuous action space in the proposed DC-PPO method, there are two ways to implement actor-critic deep neural networks, i.e., (i) two actor networks used to output two action sets simultaneously and separately and (ii) one actor network used to directly output two actions at the same time. For both two-actor and one-actor models, the state space size for each agent is the same and can be given as:

$$\begin{aligned}|s_k(t)| =& |\mathcal{I}| + |\mathcal{J}| + |C_k^{req}(t)| + |block_k(t)| + |a_k(t-1)| \\ &+ |C_k(t-1)| + |C_{sum}(t-1)| \\ =& |\mathcal{I}| + |\mathcal{J}| + 1 + 1 + 2 + 1 + 1 = |\mathcal{I}| + |\mathcal{J}| + 6.\end{aligned}$$

And the action space size for two-actor and one-actor models is also the same, calculated as $|\mathcal{I}| + |\mathcal{J}| + 1$. Then the computational complexity of the two-actor and one-actor models for each distributed agent can be calculated as (23).

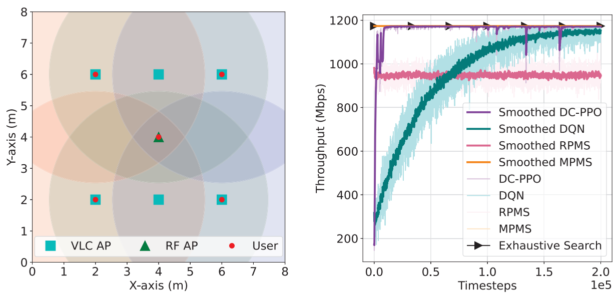
DQN: DQN includes two networks, i.e., the behavior network and the target network, therefore, the overall computational complexity is $\mathcal{O}(2(|\mathcal{I}| + |\mathcal{J}| + 6)N_H + 2N_H^2 + 2N_H(|\mathcal{I}| + |\mathcal{J}| + N_D))$.

MPMS & RPMS: For these two baseline approaches, their computations only involve choosing the AP with the highest SNR, thus resulting in the same computation complexity calculated as $\mathcal{O}((|\mathcal{I}| + |\mathcal{J}|)|K|)$.

TABLE III: DC-PPO hyperparameters

Parameter	Value
Learning rate	0.0001
Clipping coefficient (ϵ)	0.1
Discount factor (β)	0.999
Number of epoch per iteration	8
Update timestep	500

$$\begin{cases} \underbrace{\mathcal{O}((|\mathcal{I}|+|\mathcal{J}|+6)N_H + N_H^2 + N_H(|\mathcal{I}|+|\mathcal{J}|+1))}_{one\ actor} + \underbrace{\mathcal{O}((|\mathcal{I}|+|\mathcal{J}|+6)N_H + N_H^2 + N_H)}_{one\ critic} = \mathcal{O}(2(|\mathcal{I}|+|\mathcal{J}|+6)N_H + 2N_H^2 + N_H(|\mathcal{I}|+|\mathcal{J}|+2)) \\ \underbrace{\mathcal{O}(2(|\mathcal{I}|+|\mathcal{J}|+6)N_H + 2N_H^2 + N_H(|\mathcal{I}|+|\mathcal{J}|+1))}_{two\ actors} + \underbrace{\mathcal{O}((|\mathcal{I}|+|\mathcal{J}|+6)N_H + N_H^2 + N_H)}_{one\ critic} = \mathcal{O}(3(|\mathcal{I}|+|\mathcal{J}|+6)N_H + 3N_H^2 + N_H(|\mathcal{I}|+|\mathcal{J}|+2)) \end{cases} \tag{23}$$



(a) 7-AP and 5-user topology (b) Throughput comparison

Fig. 4: 5 Static users without blockage.

In summary, the MPMS and RPMS methods have a lower computational complexity compared to the two RL methods, i.e., DC-PPO and DQN. And the adopted one-actor DC-PPO has a lower computational complexity compared with DQN. Their resulting performance is shown in the following section.

D. Simulation Results

In this section, we will evaluate the performance of the DC-PPO in terms of convergence, scalability, and robustness by comparing it with the three baseline methods, i.e., DQN, MPMS, and RPMS.

Optimality and Convergence: We first evaluate the optimality and convergence speed of the proposed DC-PPO method in a special network topology as defined in Fig. 4(a). In this topology, 5 static users, 6 VLC APs and 1 RF AP are deployed, where 4 users are allocated exactly under the 4 VLC APs located at the four outer corners, and the remaining 1 user is set under the RF AP. The circled areas with different colors indicate the coverage areas of different VLC APs. In this special networking topology, the optimal solution can be easily observed. To avoid interference and achieve the optimum, the 4 users at the corners should connect to the VLC APs right above them, while the remaining 1 user should select the RF AP. In addition, all users should request the available maximum transmission power from the associated APs. This optimal solution is also confirmed by the exhaustive search method[1]. We also compare with the other three baseline approaches and the results are shown in Fig. 4(b). To better illustrate the convergence trend, an exponential moving average is applied to the raw results to obtain more smoothed results. We can see that MPMS achieves the global optimal throughput, matching the performance of the exhaustive search method. RPMS achieves the lowest sum network throughput. Even with limited information exchange, the proposed DC-PPO method enables each agent to cooperate effectively and learn a policy that results in the global optimal user association and power allocation in a short time span of 10^4 timesteps. Although DQN can also nearly achieve the optimum, its convergence speed is significantly slower.

[1]Here we use exhaustive search because it is only executed one time in the static networking scenario compared to the mobile scenario, thus our personal computer can finish it within a reasonable time.

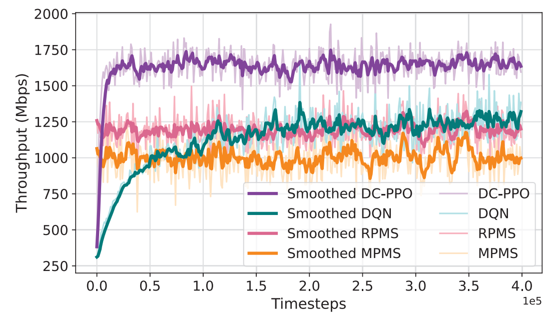


Fig. 5: Network throughput comparison for 11 mobile users with 20% blockage.

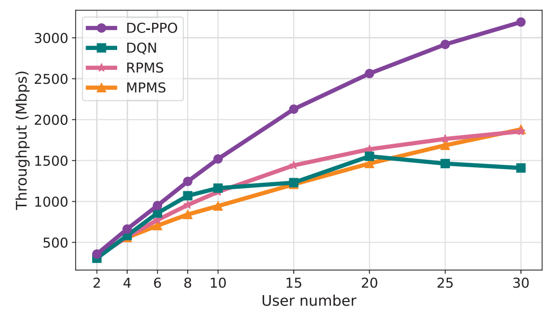


Fig. 6: Network throughput vs. user number with 20% blockage.

Furthermore, we compare the convergence speed and optimality with the same AP deployment as the above-mentioned but with an arbitrary number of mobile users and blockage probability $\mathrm{Pr}_{th} = 0.2$. Exhaustive search will not be used since it is not applicable to the dynamic networking scenarios. The results for 11 mobile users are presented in Fig. 5. Our approach not only outperforms DQN by up to 38.4% in terms of network throughput, but also converges approximately 75.25% faster. The DQN approach is comparable in performance to the RPMS method. However, the MPMS approach has the poorest performance, mainly due to the increased network interference caused by the increased number of users. This results in poorer user-AP association based on SNR obtained from the max-power strategy.

Scalability: We then investigate the scalability of our proposed DC-PPO by increasing the number of mobile users from 2 to 30 with the same AP deployment as in Fig. 4(a), where blockage probability $\mathrm{Pr}_{th} = 0.2$. It is noteworthy that the proposed distributed algorithm ensures that the neural network training process remains computationally efficient regardless of the number of users because of the distributed feature of DC-PPO.

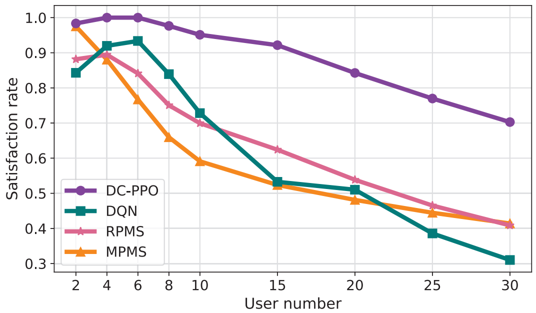


Fig. 7: Satisfaction rate vs. user number with 20% blockage.

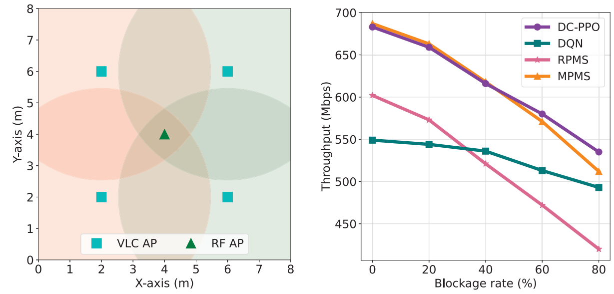
Figure 6 shows the network throughput of the proposed DC-PPO and three baseline methods for different numbers of users. For the 2-user scenario, all methods achieve comparable throughput. As the number of users increases to 4, 6, and 8, the proposed DC-PPO method achieves better throughput than the MPMS, RPMS, and DQN methods. As the user number keeps increasing larger, the performance of the DQN becomes unstable and even degrades, while the performance of the DC-PPO algorithm remains unaffected. Notably, the improvement achieved by DC-PPO over MPMS, RPMS, and DQN methods continues to widen, where it outperforms DQN, RPMS, and MPMS by approximately 127%, 72%, and 70% in the 30-user scenario. This can be explained by the fact that the MPMS and RPMS methods select the highest SNR AP for user association, regardless of the capacity limitations. In contrast, the DC-PPO can analyze capacity limitations by analyzing achievable data rate and maximum network throughput feedback. Although the DQN algorithm also uses feedback-based analysis, its performance degrades as the number of users increases, which suggests that the DQN algorithm may not be suitable for large-scale multi-agent schemes in dynamic environments.

Besides the network throughput, the user satisfaction rate is another important QoS, which is defined as:

$$\eta = \min\{1, \frac{C_{sum}}{\sum_{k=1}^{K} C_k^{req}}\}. \tag{24}$$

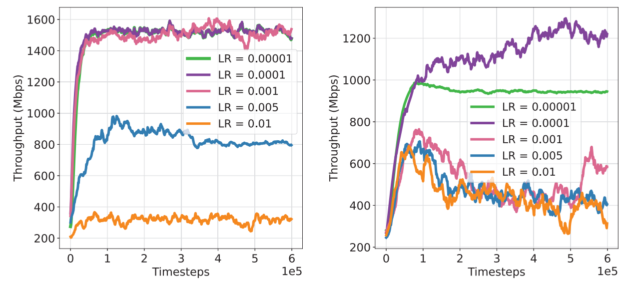
We then examine the satisfaction rate performance of the proposed approach. From Fig. 7, we can clearly observe that the proposed DC-PPO method outperforms the three baseline methods. Despite a decrease in satisfaction rate as the number of users increases, the DC-PPO method can still achieve a 70% satisfaction rate in the 30-user scenario, which is significantly higher than 31%, 40%, 41% obtained by DQN, RPMS, and MPMS, respectively.

Robustness: We have demonstrated the superior performance of DC-PPO in terms of optimality and convergence, and scalability in dynamic scenarios with various numbers of mobile users and 20% blockage probability. We then first validate the robustness of the proposed DC-PPO with respect to different levels of blockage in the dynamic indoor environment. To highlight the impact of the blockage, we adopt a sparser deployment of VLC APs with 4 VLC APs and 1 RF AP as shown in Fig. 8(a), where the VLC overlap area is smaller, implying that the probability of them being able to switch to another VLC AP is reduced when the current connection is blocked. The comparison results for the 6-mobile-user scenario are shown in Fig. 8(b). As expected, as the blockage rate increases from 0% to 80%, network throughput decreases for all methods. Specifically, the throughput decreases up to 27.7%, 11.4%, 34.2%, and 43.4% for DC-PPO, DQN, MPMS, and RPMS respectively. It is also observed that DC-PPO achieves comparable performance to MPMS When the blockage rate is low. However, as the blockage rate increases, the proposed method outperforms all three baselines.



(a) 5-AP network topology (b) Throughput vs. blockage rate

Fig. 8: 6 mobile users with varying blockage.



(a) DC-PPO (b) DQN

Fig. 9: LR vs. throughput for 10 mobile users with 20% blockage.

Next, we examine the robustness of DC-PPO in terms of learning rate (LR) compared with DQN. From Fig. 9(a) and Fig. 9(b), we can obviously see that DC-PPO not only has faster convergence speed and higher sum throughput but also is more robust to the changes of LR compared with DQN. Specifically, the performance of DC-PPO does not degrade as LR ranges from 0.00001 to 0.001, while the performance of DQN significantly drops.

VII. Conclusions

This paper studies and proposes a joint user association and power allocation problem in an indoor hybrid VLC/RF network by jointly considering user mobility, time-varying blockage and interference in VLC networks, and capacity limitations. To solve the resulting MINLP problem, we then propose a novel distributed cooperative PPO method that models each user as an agent to select optimal action from hybrid discrete user association and continuous power allocation action space in a distributed manner. Comprehensive simulation results show that our proposed method can achieve the global optimum and nearly global optimum in static scenarios and mobile scenarios with blockage, respectively, and significantly outperform the state-of-the-art methods in terms of computation complexity, convergence speed, scalability, and robustness.

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